Installing DunesForge

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Installation takes about two minutes. GPU acceleration is one extra click, and the add-on works without it.

> **A note on the name.** The product is **DunesForge**. Inside Blender the add-on's panel is still labelled **TyDunes** - that is the tab to look for under **[N]**, and the file you install is `tydunes_blender.zip`. The interface will catch up with the name; nothing about how it works changes.

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1. What you need

Blender	4.2 or newer (tested through 5.3)
OS	Windows or Linux
GPU	Optional. An NVIDIA CUDA card makes it interactive; without one it still runs
Disk	About 500 MB - most of that is the bundled texture library

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2. Install the add-on

1. Open Blender.

2. Edit > Preferences > Get Extensions (or **Add-ons** on 4.2).

3. Click the **v** arrow at the top right of that window and choose

Install from Disk...

4. Pick `tydunes_blender.zip` - the file you downloaded. Do not unzip it

first; Blender wants the zip.

5. Wait. It is a large file and Blender copies the whole texture library, so a

few seconds of an unresponsive window is normal.

6. The add-on appears in the list and is already enabled.

Where it lives now. Press **[N]** in the 3D viewport to open the side panel, then pick the **TyDunes** tab. That tab is the whole product.

> **If you do not see the tab:** your mouse must be over the 3D viewport when you press **[N]**, not over the outliner or the timeline.

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3. Turn on GPU acceleration (recommended)

DunesForge computes on the GPU through a library called **CuPy**. Blender does not ship it, so the add-on installs it for you.

1. Open the **DunesForge** panel (**[N]** > DunesForge).

2. Near the top you will see a notice that the GPU backend is missing, with a

button: **Install CuPy GPU Backend**.

3. Click it. It downloads and installs CuPy into Blender's own Python - it does

not touch any other Python on your machine.

4. Restart Blender when it finishes.

That is all. After the restart the notice is gone and the solver is on the GPU.

If the button fails

The most common reasons, in order:

Symptom	Cause	Fix
Nothing happens, or an error about permissions	Blender cannot write into its own folder	Run Blender once as administrator, click the button, then go back to running it normally
Download fails	No internet, or a corporate proxy/firewall	Install manually - see below
Installs, but the notice stays after restart	Your GPU is not NVIDIA/CUDA	Nothing to fix; use the CPU path (see §4)

Manual install. Find Blender's Python executable - it sits in `.../Blender Foundation/Blender <version>/<version>/python/bin/` - and run:

```
python.exe -m pip install cupy-cuda12x
```

Use `cupy-cuda11x` instead if your driver is CUDA 11.

Checking that it worked

In the TyDunes panel, use **GPU Diagnostics** (magnifier icon). It reports which backend is active and what your card is. If it says CuPy and names your GPU, you are done.

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4. No NVIDIA card? Read this

DunesForge runs the **same simulation** on the CPU through NumPy. Nothing is disabled, no feature is missing and the results are the same physics - it is simply slower, and how much slower depends on your grid resolution.

If you are on the CPU path:

leaving Play running.

- Start at a **lower Grid Resolution** (the **Performance** panel).
- Use **Run Steps** to advance the simulation in deliberate batches rather than
- Everything in this documentation still applies to you.

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5. Updating to a newer version

Install the new zip exactly the way you installed the first one. Blender replaces the old version.

Your scenes are safe. Every physical feature in DunesForge is off by default and does nothing at its default value, so a scene saved with an older version opens with the same numbers and simulates the same way.

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6. Uninstalling

Edit > Preferences > Get Extensions, find DunesForge, and use the **v** menu > **Uninstall**. Scenes you already built keep their meshes - those are ordinary Blender objects and do not need the add-on to render.

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7. Something is wrong and this page did not cover it

The **Diagnostics** button in the TyDunes panel opens the folder holding the diagnostic log. That file is what to send when asking for help; it records what the solver actually did rather than what it was asked to do.

Next: **QUICKSTART.md** - your first dune in five minutes.